

Research Paper 1 - Estimation of Soil Moisture in Paddy Field Using Artificial Neural Networks

Original Research Paper Link -

https://www.researchgate.net/publication/235891972_Estimation_of_soil_moisture_in_paddy_field_using_Artificial_Neural_Networks

My Google Drive Link -

https://drive.google.com/file/d/14nqXHfXogtkoq4K1T4EqZiXIMT7odQoK/view?usp=drive_link

Summary of the Paper

- **Authors:** Chusnul Arif et al.
- **Title:** Estimation of Soil Moisture in Paddy Field Using Artificial Neural Networks
- **Approach:** The paper proposes two ANN models:
 1. To estimate reference evapotranspiration (ET_o) using temperature inputs.
 2. To estimate soil moisture using ET_o, precipitation, and crop coefficient.
- **Data Limitations:** The study focuses on limited meteorological data due to cost and accessibility challenges.
- **Validation:** Model achieved $R^2 = 0.80$ (training), $R^2 = 0.73$ (validation) using historical datasets.

Identified Technological Gaps

Area	Existing Paper	Your Proposed System	Technological Gap
Data Sources	Uses only temperature and precipitation data for soil moisture prediction.	Uses real-time sensor data (soil moisture, tank water level) and weather forecasts (rain prediction).	Lack of real-time sensor integration and proactive prediction using weather forecasts.
Control Mechanism	No implementation of irrigation control—only prediction.	Automates irrigation by opening a water gate using motorized control based on predictions.	Absence of automated actuation mechanism for irrigation in the paper.

Scope	Focused on soil moisture estimation only.	Combines prediction with action (decision-making + irrigation execution).	Paper does not close the loop from estimation to automated water delivery.
Intelligence	Uses simple feedforward ANN with limited inputs and fixed parameters.	Can leverage advanced AI models (like RNNs, LSTMs, or hybrid models) to learn from temporal patterns and forecast more accurately.	Lacks exploration of temporal learning models and AI adaptation over time.
Infrastructure	Requires human effort to interpret model outputs and act accordingly.	Designed for autonomous system with minimal human intervention.	Gap in building an end-to-end autonomous solution.

Value Addition in my FYP

My Final Year Project **extends the capabilities** of existing research in these ways:

- Integrates **sensor networks** for real-time environmental monitoring.
- Uses **predictive modeling + future weather forecasts** to adjust irrigation needs dynamically.
- Introduces **motorized water gate control** for automatic irrigation—making it a **closed-loop smart farming solution**.
- Can improve water use efficiency in paddy cultivation, especially relevant for regions with water scarcity.

Research Paper 2 - Generating Soil Salinity, Soil Moisture, Soil pH from Satellite Imagery and Its Analysis

Original Research Paper Link -

https://www.researchgate.net/publication/335358334_Generating_Soil_Salinity_Soil_Moisture_Soil_pH_from_Satellite_Imagery_and_Its_Analysis

My Google Drive Link -

https://drive.google.com/file/d/1N5coe3s8d49MikTWiKb2n46sAZFQQrEs/view?usp=drive_link

Summary of the Research Paper

- **Authors:** Ghazali et al.
- **Title:** Generating Soil Salinity, Soil Moisture, Soil pH from Satellite Imagery and Its Analysis
- **Approach:** Uses **Landsat 8 satellite imagery** combined with lab/field data to estimate:
 - Soil moisture (SMI),
 - Soil salinity (SSI),
 - Soil pH (SpHI)
- **Models Used:** Regression-based “**paddy leaf**” and “**bare soil**” models
- **Phenological Consideration:** Soil parameters monitored across pre-harvest and post-harvest phases.
- **Accuracy:** Achieved RMSE = 1.40 and PRMSE = 24%.

Identified Technological Gaps

Area	Satellite-Based Model (Ghazali et al.)	Your Proposed Smart Gate System	Technological Gap
Data Source	Relies on multi-temporal satellite imagery and field survey data	Uses real-time on-ground sensors (moisture, water level, weather forecast)	Satellite imagery is not real-time, limited by revisit cycles and cloud cover
Irrigation Control	Purely observational and analytical— no control action	Introduces motorized gate control that responds to predictions	No feedback/actuation mechanism exists in the paper
Granularity	Resolution is coarse (30m ² per pixel)	Targets local, fine-grained control using ground-level sensors	Remote sensing can't support plot-level decision-making reliably
Prediction Use Case	Uses SMI/SSI/SpHI for monitoring and mapping	Uses sensor + forecast data to predict and act on future needs	Gap in integrating predictive irrigation decision logic
Adaptability	Static regression models, not real-time adaptable	Capable of learning continuously and adjusting dynamically (e.g., LSTM models)	Models are not adaptive to dynamic field conditions

Value Addition in my Smart Water Gate FYP

My system improves on the satellite-based monitoring approach by:

- Enabling **real-time control** through physical gate actuation.
- Utilizing **fine-resolution sensor data** (soil moisture, water level).
- Including **future weather conditions (rain prediction)** into the logic.
- Designing an **end-to-end closed-loop solution**: monitor → predict → act.

Research Paper 3 - IoT based Smart Irrigation System using Machine Learning Models

Original Research Paper Link -

<https://www.sciencedirect.com/science/article/pii/S0957417422009563>

My Google Drive Link -

https://drive.google.com/file/d/1wzw1K3nR9RBZgW3HETRYYY3cfkxsddc4n/view?usp=drive_link

Summary of the Research Paper

- **Authors:** P. Raja et al.
- **Title:** *IoT based Smart Irrigation System using Machine Learning Models*
- **Published In:** Expert Systems with Applications, Elsevier (2022)
- **Core Features:**
 - Combines **IoT devices** (sensors for soil moisture, humidity, temperature).
 - Employs **machine learning models** (Random Forest, SVM, Naive Bayes, XGBoost) to predict irrigation needs.
 - Uses **weather APIs** to include environmental context.
 - Sends irrigation decisions to a **relay circuit** (solenoid valve for irrigation).
 - Achieves 95.6% prediction accuracy with RF.

Identified Technological Gaps

Area	Existing IoT System (Raja et al.)	Your Proposed System	Technological Gap
ML Models	Uses general-purpose ML models (RF, SVM, etc.) for classification	Can explore temporal models (LSTM/RNN) or hybrid decision logic for better time-series irrigation prediction	Existing models may lack temporal insight needed for dynamic soil conditions
Control	Uses relay switch to control solenoid valves	Uses motorized water gate mechanism —potentially more robust for paddy field irrigation	Paper focuses on pipe irrigation , not suitable for field-level water gate systems
Environmental Inputs	Includes weather forecasts, temperature, humidity, and soil moisture	Adds tank water level sensor and focuses specifically on rain prediction and water availability	Lacks water source feedback (e.g., tank monitoring) in the irrigation loop
Crops and Context	General applicability, no specific tuning for paddy fields	Tailored for paddy cultivation , which has unique waterlogging and drainage needs	No crop-specific logic or constraints modeled in the referenced system
Automation Design	Basic control via relay, focuses mostly on decision making	Full integration of control and mechanical automation (gate operation, water release control)	Your system introduces electro-mechanical integration , which is missing in the paper
Adaptability & Customization	No adaptive control logic across growing stages or changing water availability	Includes potential to adapt to crop growth stages, forecast changes, and tank levels	Static rule application; lacks adaptive irrigation policy formulation

Value Addition in my Smart Water Gate FYP

My project advances the state-of-the-art by:

- Implementing **smart gate actuation** for paddy-specific needs rather than generic irrigation valves.
- Considering **water tank levels**, making it more sustainable and resource-conscious.
- Using **weather forecast (especially rain prediction)** for proactive irrigation control.

- Opening opportunities for **long-term deployment** with learning-based control systems (e.g., reinforcement learning or LSTM-based predictions).
- Focusing on **real-time irrigation for paddy**, which requires more nuanced water handling compared to general horticulture.

Research Paper 4 - Estimating Soil Moisture Condition of Paddy Fields by Using Optical Remote Sensing Imagery

Original Research Paper Link -

<https://iopscience.iop.org/article/10.1088/1755-1315/1109/1/012067>

My Google Drive Link -

https://drive.google.com/file/d/1zjxAclg9nUwY0GesJE7oSeI2rmLCraIB/view?usp=drive_link

Summary of the Research Paper

- **Authors:** Rizqi I'anatus Sholihah et al.
- **Conference:** IOP Conf. Ser.: Earth and Environmental Science, 2022
- **Focus:** Used **Landsat 8 OLI/TIRS data** to estimate:
 - Soil Moisture Index (SMI)
 - Vegetation Index (NDVI)
 - Land Surface Temperature (LST)
- **Methodology:**
 - Derived LST from thermal infrared data.
 - Used NDVI and LST to calculate SMI for paddy fields in East Java, Indonesia.
 - Mapped spatial distribution of soil moisture using satellite imagery.

Identified Technological Gaps

Area	Remote Sensing Model (Sholihah et al.)	Your Proposed Smart System	Technological Gap
Data Type	Optical satellite imagery (Landsat 8)	Real-time ground sensors (moisture, tank level, temperature forecast)	Lack of real-time and on-field granularity in the paper
Purpose	Soil moisture monitoring and mapping	Predictive model to act and irrigate accordingly	Monitoring only; lacks automated response system

Spatial Resolution	Medium spatial resolution (30m), passive sensing	Local plot-level control with precise irrigation via gate control	Can't be used for micro-level irrigation decisions
Environmental Integration	Uses NDVI and LST as indirect proxies	Includes direct sensing + future rain prediction for accurate irrigation timing	No weather or climate forecast integration
Temporal Frequency	Periodic imagery based on satellite overpass (8–16 days)	Near real-time decision-making based on sensor input	Temporal gap leads to delayed insight , not suitable for dynamic control
Irrigation Feedback	No decision-making or control systems	Fully integrates automated gate with motorized control	No feedback loop from monitoring to execution in the paper

Value Addition in my Smart Water Gate FYP

My system improves on satellite-based monitoring by:

- **Closing the loop** from prediction to real-world **irrigation control**.
- **Real-time adaptability**, responding to immediate field conditions (moisture and tank level sensors) and forecasts.
- Using **fine-grained data** from on-site sensors suitable for paddy fields.
- Enabling **automated control of irrigation infrastructure** with motorized gates.
- Tailoring for **paddy-specific requirements**, which satellite imagery cannot address directly.

Research Paper 5 - A low-cost approach for soil moisture prediction using edge computing devices and machine learning

Original Research Paper Link -

<https://www.sciencedirect.com/science/article/pii/S0048969722021593>

My Google Drive Link –

https://drive.google.com/file/d/1DdFd1Kwa8rFD2hmeK-T5B86jbc-p9vwx/view?usp=drive_link

Summary of the Research Paper

- **Authors:** G. Ramesh Kumar et al.
- **Published In:** *Science of the Total Environment*, 2022
- **Objective:** Design a **low-cost, edge-based** solution for soil moisture prediction.
- **Method:**

- Utilizes **on-field sensors** for soil moisture, temperature, humidity, etc.
- Performs **prediction using ML models (Random Forest, ANN)** directly on **edge devices** (e.g., Raspberry Pi).
- Focused on **reducing latency and dependency on cloud systems**.
- **Key Contributions:**
 - Real-time inference on edge devices.
 - Promotes cost-effectiveness for small-scale farming.
 - Achieves good accuracy with ML models, emphasizing **affordability and independence**.

Identified Technological Gaps

Area	Edge-Based Prediction (Ramesh Kumar et al.)	Your Proposed System	Technological Gap
Prediction Platform	Uses ML models deployed on edge devices	Could adopt AI models that also consider forecast and real-time irrigation control	Focuses on prediction only, not on integration with mechanical irrigation
Control Mechanism	No control element—no irrigation action triggered	Introduces motorized water gate that opens automatically	Missing the actuation part of the feedback loop
Environmental Context	Uses sensor data like soil temperature and humidity	Adds tank water level and rain prediction , making it more context-aware	Rain forecast & water availability not integrated in their model
System Scope	Soil moisture prediction only; assumes human intervention post-prediction	Fully autonomous irrigation system that requires no human input post-deployment	No end-to-end automation in the proposed system
Target Crop	General agriculture (not crop-specific)	Designed for paddy fields , which have unique water needs (flooding, drainage)	Lacks crop-specific optimization , which is key in your FYP
Hardware Focus	Focuses on affordable ML deployment via edge	Also includes electromechanical integration for water control	Paper does not address hardware actuation or water

			infrastructure management
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Value Addition in My Smart Water Gate FYP

My system takes the core idea of edge-based prediction further by:

- **Integrating mechanical actuation** (smart gate control), turning prediction into **action**.
- Including **weather forecast and water tank level monitoring** to optimize water use.
- Designing for **paddy-specific scenarios**, addressing irrigation peculiarities like field flooding.
- Maintaining **autonomous operations** with little to no farmer intervention.
- Potentially blending **on-device inference** with cloud support for long-term learning.

Research Paper 6 - Soil moisture quantity prediction using optimized machine learning algorithm

Original Research Paper Link -

<https://www.sciencedirect.com/science/article/pii/S2210537918302464>

My Google Drive Link -

[https://drive.google.com/file/d/1uXyVJdG8tWGFabK3n3-FzW12v5RYIQ7/view?usp=drive link](https://drive.google.com/file/d/1uXyVJdG8tWGFabK3n3-FzW12v5RYIQ7/view?usp=drive_link)

Summary of the Research Paper

- **Authors:** Nirala et al.
- **Published In:** *Sustainable Computing: Informatics and Systems* (2020)
- **Focus:** Predicting **quantitative soil moisture** levels using optimized ML techniques.
- **Methodology:**
 - Applies several ML models: SVM, ANN, Random Forest, and ensemble classifiers.
 - Uses **optimization techniques** (e.g., firefly algorithm) to tune model parameters.
 - Data collected from **weather stations and field sensors** (temperature, humidity, etc.).
- **Goal:** Improve **accuracy and generalization** of moisture prediction systems for precision agriculture.
- **Result:** ANN with Firefly optimization yielded the best performance (98% accuracy).

Identified Technological Gaps

Area	Optimized ML Approach (Nirala et al.)	Your Proposed Smart System	Technological Gap
Purpose	Focuses on improving prediction accuracy through algorithmic tuning	Focuses on closing the loop from prediction to actuation	Paper does not progress to real-world irrigation actions
Environmental Inputs	Uses standard sensor inputs (humidity, temperature)	Incorporates rain prediction and tank water levels for proactive irrigation	No forecasting or water availability logic present
Control Element	Predicts soil moisture; assumes external action by farmers	Automates the control of a water gate using motor mechanisms	No hardware integration or field deployment planning
System Autonomy	Semi-automated; still requires manual action post-prediction	Fully autonomous—runs end-to-end with minimal human input	No transition from smart analytics to smart action in their system
Crop-Specific Focus	General application; not specific to paddy or waterlogged fields	Designed for paddy field conditions , considering unique water requirements	Lacks crop-specific customization or adaptation logic
Temporal Dynamics	Models trained on static datasets	Can integrate time-series forecasting using models like LSTM	Lacks temporal learning and adaptation to field changes

Value Addition in My Smart Water Gate FYP

My project builds on this predictive work by:

- Applying **real-time predictions** to a **motorized water gate** for paddy-specific irrigation.
- Leveraging **additional real-world variables** like tank level and rain forecasts to improve decision-making.
- Enabling **temporal forecasting** to adjust irrigation based on weather trends and crop stages.

- Closing the **AI–mechanical action loop**, offering a practical solution for water conservation.
- Designing for **paddy-specific scenarios**, addressing irrigation peculiarities like field flooding.

Research Paper 7 - Soil moisture quantity prediction using optimized machine learning algorithm

Original Research Paper Link -

<https://www.sciencedirect.com/science/article/pii/S0016706117321031>

My Google Drive Link -

https://drive.google.com/file/d/1huQcC0gNh5N65YFNx3Ds4FTTrvNGJG4fF/view?usp=drive_link

Summary of the Research Paper

- **Authors:** Shahaboddin Shamshirband et al.
- **Published In:** *Geoderma*, 2018
- **Focus:** Short-term **soil moisture forecasting** using a **hybrid ML model**.
- **Technique:**
 - Combines **M5 model tree** (a rule-based regression model) with the **firefly optimization algorithm**.
 - Uses meteorological inputs: rainfall, temperature, evaporation, humidity.
 - Forecasts soil moisture over short horizons with high accuracy.
- **Performance:** Achieved RMSE and MAE values superior to standalone models (e.g., ANN, RF, SVM).

Identified Technological Gaps

Area	Hybrid Forecasting Model (Shamshirband et al.)	Your Proposed System	Technological Gap
Prediction Technique	Uses hybrid ML (M5 + firefly) for time-series soil moisture forecasting	Can use LSTM, ensemble or RL-based models for real-time decision-making	No neural or adaptive learning for irrigation control

Actuation Layer	Purely predictive—no control or automation component	Uses motorized gate to act on predictions and control irrigation	Missing physical irrigation implementation entirely
Input Variables	Rainfall, temperature, humidity, evaporation	Adds tank water level and real-time soil moisture sensors	No integration of water source conditions or sensor fusion
System Output	Provides numerical soil moisture forecasts	Uses predictions to automatically manage irrigation	Forecast output not connected to practical execution
Field Realism	Model tested on existing weather station data	Designed for real deployment in paddy fields , including rainwater management	No field integration strategy mentioned
Time-Awareness	Forecasts short-term conditions (24–48h)	Adjusts dynamically based on live weather updates + future rain prediction	Model lacks forecast update adaptability during real-time deployment

Value Addition in My Smart Water Gate FYP

My system:

- Applies predictions directly into **practical irrigation systems** (gate-based control).
- **Fuses real-time sensors (e.g.- tank level) with forecasting data (e.g.-rain forecast)**, enabling intelligent decision-making.
- Goes beyond numerical modeling by delivering an **end-to-end automated solution** for paddy irrigation.
- Supports **fine-grained control** tailored to paddy field requirements, not just generic soil moisture estimation.

Research Paper 8 - Deep learning for proximal soil sensor development: A survey

Original Research Paper Link -

<https://www.sciencedirect.com/science/article/pii/S095741742200269X>

My Google Drive Link -

https://drive.google.com/file/d/1EyppEiKk86mwOdT1DlyenwjcFidrcDJg/view?usp=drive_link

Summary of the Research Paper

- **Authors:** Zhang et al.
- **Published In:** *Expert Systems with Applications*, 2022
- **Purpose:** Survey of **deep learning applications** for **proximal soil sensing**.
- **Scope:**
 - Focuses on **developing and optimizing soil sensors** using DL.
 - Covers applications like **soil moisture, salinity, nitrogen** detection via:
 - Spectral data (e.g., NIR)
 - Imaging
 - Sensor fusion techniques
 - Discusses deep networks: CNNs, autoencoders, LSTM for sensor calibration and improvement.
- **Key Insight:** DL enhances **sensor data interpretation**, model accuracy, and calibration—but systems are **largely diagnostic**, not autonomous.

Identified Technological Gaps

Area	DL-Based Sensor Development (Zhang et al.)	Your Proposed Smart System	Technological Gap
Main Focus	Improve sensor precision using deep learning (via RGB image classification)	Build a smart irrigation system using real-time multi-sensor inputs and predictions	Focuses on sensor development and classification, not predictive decision-making or control
System Design	Concerned with data classification using visual features (image textures)	Incorporates AI prediction + mechanical actuation (motorized water gate)	No integration with physical irrigation hardware or multi-sensor decision systems
Control Layer	No mention of gate control, irrigation infrastructure , or field execution	Automates water flow via real-time logic and physical gate actuation	No automation or control pipeline to close the feedback loop
Prediction Usage	DL improves soil moisture class recognition (classification , not forecasting)	Uses ML to predict future irrigation needs (forecasting + regression models)	No time-based forecasting or future moisture requirement prediction is done
Deployment Strategy	Built for controlled conditions (static camera + pot soil images); not field-tested	Field-ready , using low-cost sensors, controllers, and adaptive prediction models	Not validated in dynamic, real-world field environments

Real-time Automation	Only offline training and classification ; no real-time control system discussed	Designed for autonomous, closed-loop real-time irrigation	No real-time control or feedback — remains a sensing system only
Environmental Adaptability	Limited to fixed lighting and fixed soil pot setups	Built to work across changing field, weather, and soil conditions	No demonstration of adaptability to field lighting, weather variability , or multiple crops
Scalability	Model limited to image input from a few soil types	Flexible system with scope for multi-sensor scaling and farm-wide deployment	No plan or method for scaling the model across larger or multiple fields

Value Addition in My Smart Water Gate FYP

My system builds upon this sensing research by:

- Using **multiple sensor inputs (soil, tank level, rain forecast)** for real-world irrigation decision-making instead of Camera-based vision sensors.
- Connecting **AI-powered analysis** directly to **field actuation** (e.g., opening the gate).
- Enabling **resource-aware, rain-forecast-based irrigation** tailored to paddy field constraints.
- Designing an operational system that is **deployable**, not just experimental.

Research Paper 9 - Estimation of root-zone soil moisture from ground and remote sensing data using machine learning

Original Research Paper Link -

<https://www.sciencedirect.com/science/article/pii/S0034425721001528>

My Google Drive Link -

https://drive.google.com/file/d/12A9TAk2mSuKDG6rPfKsNdoTjdwTFAC75/view?usp=drive_link

Summary

- **Authors:** Babaeian et al.
- **Published In:** Remote Sensing of Environment, 2021
- **Focus:**
 - Estimating **Root Zone Soil Moisture (RZSM)** using:
 - **High-resolution UAV multispectral imagery (VIS, NIR)**
 - **Soil physical and hydraulic properties**
 - **Automated Machine Learning (AutoML)** with multiple models (NN, DRF, GBM, GLM)
- **Methodology:**
 - Created 12 hierarchical ML models combining **vegetation indices (NDVI, NTR)** and **soil data** (texture, bulk density, porosity, field capacity, etc.)
 - Evaluated accuracy using TDR sensor ground-truth data at 2, 10, and 50 cm depths.
- **Results:**
 - Best RZSM performance when near-surface soil moisture was included in models.
 - $RMSE < 0.02 \text{ cm}^3/\text{cm}^3$ and $R^2 > 0.90$ for best-performing models.
 - NDVI alone was insufficient; soil attributes were essential for deeper prediction accuracy.
- **Key Contribution:**
 - Demonstrated the power of **AutoML** combined with **soil-sensor fusion** for accurate field-scale moisture mapping.

Identified Technological Gaps

Area	Root-Zone ML Estimation (Srivastava et al.)	Your Proposed Smart System	Technological Gap
Estimation Target	Focused on root-zone soil moisture prediction	Uses surface moisture + forecast + water availability for irrigation decision-making	Model limited to prediction only , not irrigation execution
Control Mechanism	No irrigation or actuation system considered	Automates water gate control using motorized mechanism	No physical integration or feedback control
Data Source	Remote sensing + ground-based for model training	Live sensor data + rain forecast + tank water level	Model trained on past data, not designed for real-time adaptation
Prediction Purpose	Improve RZSM estimation accuracy for monitoring	Predict when and how much to irrigate , then act	Focuses on insight only, not operational use

Sensor System	Relies on external satellite + sparse ground sensors	Uses in-field sensors for real-time feedback	Satellite data limits temporal and spatial resolution
Crop Specificity	General agricultural use	Designed specifically for paddy fields	No tailoring to flood-irrigation crops like paddy

Value Addition in My Smart Water Gate FYP

My project distinguishes itself by:

- Integrating **real-time sensing and actuation**, not just estimation.
- Being **designed for paddy fields**, which require different irrigation strategies than root-zone-based crops.
- Creating a **feedback-enabled closed-loop system** from prediction to gate control.
- Incorporating **forecasting and water source status** (tank level), which this paper does not consider.

Research Paper 10 - Multi-layer high-resolution soil moisture estimation using ensemble learning and spatiotemporal fusion

Original Research Paper Link -

<https://www.sciencedirect.com/science/article/pii/S0034425721004260>

My Google Drive Link –

https://drive.google.com/file/d/1DjNO-sZZRH54MCGd9xMMJxy-akQRL5It/view?usp=drive_link

Summary

- **Authors:** Srivastava et al.
- **Published In:** Remote Sensing, 2021
- **Focus:**
 - Predicting **Root-Zone Soil Moisture (RZSM)** using:
 - **Surface Soil Moisture (SSM)** data from ESA-CCI and SMAP satellites
 - **Vegetation indices (NDVI)**
 - **Meteorological data**

- **Machine Learning models** (Random Forest, Support Vector Regression, Artificial Neural Network)
- **Goal:**
 - Bridge the gap between **surface** and **root-zone** soil moisture using **hybrid data-driven modeling**.
- **Methodology:**
 - Trained ML models on SSM + auxiliary inputs to predict soil moisture at **multiple depths (0–100 cm)**.
 - Model performance validated against **in situ soil moisture data** from SCAN and USCRN networks.
 - Used **feature importance ranking** to determine key predictors (NDVI, rainfall, air temperature).
- **Results:**
 - **Random Forest (RF)** consistently outperformed SVR and ANN.
 - Achieved **$R^2 > 0.80$** for 0–100 cm depth predictions.
 - High-resolution maps (36 km SMAP data) were generated for operational forecasting.

Identified Technological Gaps

Area	Multi-Layer Estimation (Wei et al.)	Your Proposed Smart System	Technological Gap
Focus	High-resolution multi-depth moisture estimation	Focused on surface-level control + rain + tank data for real-time irrigation	Paper prioritizes depth modeling, not irrigation use
Irrigation Control	No irrigation mechanism; only moisture mapping	Automates irrigation via motorized water gate	No integration with water distribution infrastructure
Data Integration	Uses advanced satellite + sensor fusion for spatial enhancement	Uses simple, local sensors + weather forecast for decision-making	Their system is data-heavy and requires external sources
Scalability	Research-grade system, dependent on satellite imagery	Scalable in-field system using low-cost sensors and open hardware	Approach may not be feasible for real-time control in small farms
Temporal Resolution	Improved using spatiotemporal fusion, but still satellite-constrained	Based on live sensor inputs and predictive models	Not truly real-time , thus limited for responsive irrigation
System Output	Soil maps and time-series analysis	Direct irrigation actions based on forecast and resource availability	No action layer— purely analytical outcome

Value Addition in My Smart Water Gate FYP

My system innovates beyond this work by:

- Delivering **actionable outcomes**—controlling irrigation gates, not just generating soil data.
- Relying on **affordable, deployable tech**—ideal for actual paddy farmers.
- Integrating real-time **forecast + water availability**, leading to **sustainable irrigation**.
- Focusing on **surface-level irrigation logic**, which is practical for paddy fields.

Research Paper 11 - Automatic gate controlling system for sustainable agriculture water management in Sri Lanka: A review

Original Research Paper Link -

[https://www.researchgate.net/publication/380288328 Automatic gate controlling system for sustainable agriculture water management in Sri Lanka: A review](https://www.researchgate.net/publication/380288328_Automatic_gate_controlling_system_for_sustainable_agriculture_water_management_in_Sri_Lanka_A_review)

My Google Drive Link –

[https://drive.google.com/file/d/1JCvUnvZU-JnBYNNISsC9A5uRu2qZjPo8/view?usp=drive link](https://drive.google.com/file/d/1JCvUnvZU-JnBYNNISsC9A5uRu2qZjPo8/view?usp=drive_link)

Summary

- **Authors:** Rathnayaka et al.
- **Published In:** *Precision Agriculture Science and Technology*, 2024
- **Focus:**
 - Reviews and proposes a **smart, fully automated irrigation system** using:
 - **Sensor nodes** (for water level, soil moisture, temp, humidity)
 - **Motorized sluice and feeder canal gates**
 - **Closed-loop PID controller**
 - **Web dashboard for remote monitoring**
 - Designed for paddy irrigation in Sri Lanka's dry zones to replace outdated, manual canal systems.
- **System Highlights:**
 - **Sensor modes** dynamically switch between regular and PID control.
 - **Gate components:** Reservoir gate, Main canal regulator, Feeder canal, Paddy field gate.

- **Wireless peer-to-peer ESP32 communication**, solar-powered sensors, and cloud integration.
- Emphasizes **precision water distribution, real-time feedback, and emergency manual override**.
- **Conclusion:**
 - Successfully demonstrates a **closed-loop, locally deployable, scalable** smart water gate system.
 - Addresses water scarcity, labor shortages, and sustainability.
 - One of the **most complete real-world implementations** of smart irrigation for paddy fields reviewed to date.

Identified Technological Gaps

Area	Given Paper	My Proposed Smart System	Technological Gap
Machine Learning Use	Uses fixed PID and rule-based control logic	Integrates AI/ML models to predict required soil moisture and schedule irrigation	Lacks predictive intelligence for forecast-driven irrigation
Rain Forecasting	No rain prediction or environmental forecasting integrated	Uses weather APIs or ML-based rain predictors to optimize timing	No pre-irrigation decision making based on rainfall likelihood
Irrigation Requirement Estimation	Responds to sensor values (threshold-based)	Predicts how much water is required based on trends and forecasts	Only reactive; no proactive irrigation strategy
Crop Sensitivity or Scheduling	Single-mode water level strategy (e.g., 8 cm target) for all stages	Can adapt irrigation based on crop growth stage or evapotranspiration	Lacks dynamic scheduling for crop growth stages

Value Addition in My Smart Water Gate FYP

- **Incorporates intelligent prediction models** for soil moisture and rain, enabling **proactive irrigation control** rather than reactive gate triggering.
- Goes beyond PID logic by introducing **adaptive AI-based decision-making** to fine-tune water supply based on future conditions.
- Provides a modular and extendable architecture that supports **crop-specific irrigation scheduling**, improving yield and water efficiency.
- Maintains all practical advantages of the reviewed system (sensor nodes, PID control, solar power, remote dashboards) while enhancing with **predictive capability and real-time optimization**.

- Bridges the gap between **smart sensing and smart reasoning**, offering both automation and intelligence for sustainable irrigation.

Research Paper 12 - Automated Water-Gate Controlling System for Paddy Fields

Original Research Paper Link -

https://www.researchgate.net/publication/349658847_Automated_Water-Gate_Controlling_System_for_Paddy_Fields

My Google Drive Link –

https://drive.google.com/file/d/1KcOSer-beg6FOGgovtEWghTIjnJ6ENap/view?usp=drive_link

Summary

- **Authors:** Sanjula et al.
- **Published In:** IEEE ICAC, 2020
- **Focus:**
 - Proposes an **IoT-based smart irrigation system** to automate water distribution from reservoirs to paddy fields.
 - Uses **sensor nodes**, a **paddy-field gate**, and a **reservoir gate**, all coordinated through **cloud-based control** and a **web dashboard**.
 - Introduces two control modes:
 - **Regular Mode:** Maintains water at 8 cm based on sensor thresholds.
 - **PID Mode:** Uses closed-loop control to continuously maintain target water level.
 - Predicts short-term rain using a **SVM model** trained on Colombo weather data.
 - Includes **ESP-NOW-based peer-to-peer networking**, solar power, and cloud data logging.
- **Implementation:**
 - Real-time testing conducted with variable flow rates, temperature, and set-point maintenance under PID logic.
 - Achieved accurate **8 cm water level regulation** using sensor feedback and gate height adjustment.
- **Conclusion:**
 - Offers a **complete automation system** from sensing to control for paddy irrigation using IoT and PID control. However, lacks intelligent prediction or optimization.

Identified Technological Gaps

Area	Given Paper	My Proposed Smart System	Technological Gap
Decision Logic	Rule-based with thresholds and PID	AI-based predictions using ML to determine irrigation need	Lacks intelligent prediction model for soil moisture requirement
Forecast Integration	Short-term rain prediction using SVM from static dataset	Dynamic weather forecast using API or ML + sensor fusion	Forecasting model is static and location-limited
Moisture Requirement Estimation	Responds reactively to moisture thresholds	Predicts future water need and controls gate accordingly	No proactive prediction of irrigation demand
Adaptive Control	Fixed water level (8 cm) for all crop phases	Adaptive control based on crop growth stage, soil moisture trends	No support for dynamic scheduling based on crop or field variability

Value Addition in My Smart Water Gate FYP

- Introduces **AI/ML-driven predictions** for soil moisture and rainfall, enabling proactive irrigation decisions.
- Replaces static PID logic with **adaptive models** that account for crop phase, environment, and resource optimization.
- Provides a fully **closed-loop system**: from prediction → decision → action, linking sensors to **automated gate control**.
- Enhances forecast flexibility with **real-time weather APIs or ML-based dynamic forecasting** beyond fixed-location datasets.
- Designed for **scalability and configurability**, allowing tailored irrigation strategies for different fields or crops.

Research Paper 13 - Design of Automatic Watergate System Using ESP32 Microcontroller Based on Fuzzy Logic Method

Original Research Paper Link -

<https://ieeexplore.ieee.org/document/9678410>

My Google Drive Link –

https://drive.google.com/file/d/1CcrFYqL4cI7N9TvtHoSp861f6XeesBSL/view?usp=drive_link

Summary

- **Authors:** Rama et al.
- **Published In:** IEEE, 2021
- **Focus:**
 - Designing a **fuzzy logic-based automatic watergate system** using:
 - **ESP32 microcontroller**
 - **Ultrasonic water level sensor**
 - **Soil moisture sensor**
 - **DC motor-driven mechanical gate**
 - The system determines the **gate opening level (0%, 50%, or 100%)** based on sensor input processed through a **Mamdani fuzzy inference system**.
- **System Flow:**
 - Sensor data (water level + soil moisture) → Fuzzy logic controller → Motorized gate actuation.
 - Radio modules (NRF24L01) used for **wireless communication** in remote environments.
 - MATLAB used for rule development and defuzzification logic testing.
 - Membership functions (low, normal, high) defined for both sensors; rules built for 9 possible input conditions.
 - Hardware tested with voltage calibration and signal propagation testing (with and without obstacles).
- **Key Contributions:**
 - Flexible gate control using fuzzy rules instead of binary thresholds.
 - Design adaptable to **rural/remote areas** with weak internet via radio-based communication.
 - Successfully simulated and tested with sensor accuracy and defuzzification validation.

Identified Technological Gaps

Area	Given Paper	My Proposed Smart System	Technological Gap
Decision Logic	Fuzzy logic rules based on real-time	ML/AI-based prediction of future irrigation requirement	No predictive modeling; reacts to present sensor data only

	water and soil moisture levels		
Forecast Integration	No use of weather or rain forecasts in decision making	Uses weather forecasts or ML rain predictors for planning irrigation	Cannot skip irrigation during upcoming rain events
Irrigation Scheduling	Limited to fixed output levels (0%, 50%, 100%) based on rule-based logic	Provides flexible irrigation control based on AI-estimated moisture demand	Less granular, no dynamic computation of irrigation quantity
Crop Specific Adaptation	Single logic rule set for any condition	Adaptable AI models that adjust based on crop stage and environmental changes	Fuzzy system cannot adapt to seasonal or crop-stage variation

Value Addition in My Smart Water Gate FYP

- **Integrates ML-based soil moisture forecasting**, offering proactive irrigation rather than reactive fuzzy thresholding.
- Combines fuzzy-style flexibility with **AI adaptability**, enabling system learning and evolution over time.
- Supports **external data integration (weather APIs, forecast models)** to prevent irrigation during forecasted rainfall.
- Fully modular and scalable; deployable with or without internet access using fallback communication like LoRa or offline logging.
- Optimizes water usage with **real-time, AI-informed, dynamic actuation**, beyond pre-defined fuzzy rule responses.

Research Paper 14 - Gate Guard: Streamlining Agriculture with Smart Water Gate System

Original Research Paper Link -

<https://ieeexplore.ieee.org/document/10841506>

My Google Drive Link –

https://drive.google.com/file/d/1WMRSkMOepjbzWnAwnLup_RRuWLQRcjjc/view?usp=drive_link

Summary

- **Authors:** Niharika et al.
- **Published In:** Proceedings of IEEE ICACRS, 2024
- **Focus:**
 - Proposes an **IoT-based water gate automation system** for paddy fields using:
 - **Arduino Uno** controller
 - **Water level sensor** for detecting high/low field water levels
 - **Servo motor** for automated gate opening/closing
 - **ESP8266 Wi-Fi** module to send data to **ThingSpeak cloud**
 - **GSM SIM800i** module to send SMS alerts to farmers
- **System Workflow:**
 - Water level sensor triggers Arduino logic → If threshold is exceeded, gate opens → SMS sent to farmer and data logged to cloud.
 - 16x2 LCD shows real-time field water status.
 - Farmer receives alert: *"WATER LEVEL HIGH, GATE OPENED"*.
 - System includes a web dashboard via ThingSpeak for real-time data access.
- **Key Strengths:**
 - Combines **local sensor actuation with cloud-based remote monitoring**.
 - Designed to work in **rural, low-internet environments** using GSM backup.
 - Minimizes human intervention and supports **automated drainage** to avoid waterlogging.
- **Conclusion:**
 - Presents a **functionally complete low-cost smart gate system** using basic sensors and microcontrollers.
 - However, it does not include **predictive irrigation logic**, AI-based adaptability, or advanced scheduling mechanisms.

Technological Gaps Compared to Your Proposed System

Feature	Gate Guard (2024)	Your Proposed FYP	Technological Gap
Decision Logic	Basic rule-based thresholds	AI-based prediction of required soil moisture using multi-variable inputs	No intelligent model – no learning or forecasting
Environmental Inputs	Only current water level	Uses soil moisture, tank level, and weather forecast	Lacks multi-factor integration and predictive control
Adaptability	Static threshold; manually set	Adapts to field, season, and crop stage using AI	No self-calibration or context-awareness

Sensor Coverage	One sensor input (water level)	Combines multiple sensors : soil moisture, temp, water level	Limited environmental sensing
Forecast Integration	None	Uses weather and rain forecast to optimize timing	Cannot plan ahead for future conditions
Application Domain	General irrigation fields	Optimized for paddy cultivation , including flooding control	No paddy-specific modes or stages (e.g., transplanting)
Gate Control	Opens/closes gate based on one signal	Dynamically predicts when and how much to open	Less precision and no optimization model
Data Use	Cloud storage for logging only	Cloud/local model training , adaptive feedback	Data not used for continuous model improvement

Value Addition in My Smart Water Gate FYP

- Transitions from **threshold-based** to **prediction-based** control.
- Incorporates **AI/ML** models to determine **when and how much water** is needed.
- Integrates **rain forecasts, soil data, and tank status** to optimize decisions.
- Designed for **field deployment** with real-time, low-cost smart automation.
- Provides **intelligent irrigation tailored for paddy crop cycles**.

Research Paper 15 - Design and Control System of Sluice Gate With Web-Based Information

Original Research Paper Link -

<https://ieeexplore.ieee.org/document/9650409>

My Google Drive Link –

https://drive.google.com/file/d/15J8p0nHPRMsRF4wNB_I_RgAf6WoAy7-Z/view?usp=drive_link

Summary

- **Authors:** Desnanjaya & Nugraha

- **Published In:** IEEE ICSGTEIS, 2021
- **Focus:**
 - Developed a **web-based irrigation sluice gate system** for flood-prone rain-fed rice fields in Bali.
 - Combines hardware (Arduino, ultrasonic sensor, servo motor) and software (SIM900, website dashboard).
 - **Automatically monitors water level** at the field gate (“Temuku”) and allows farmers to control it **remotely** via a web interface or SMS.
- **System Components:**
 - **Arduino Uno** – main processor
 - **Ultrasonic HC-SR04** – water level sensor
 - **SIM900 module** – sends sensor data to database and receives SMS commands
 - **Servo motor** – controls gate opening/closing
 - **LCD + LED** – displays local water level status
- **User Roles:**
 - **General users** – view water levels
 - **Pekaseh (subak leader)** – manage accounts
 - **Farmers** – control sluice gates via dashboard (open/half-open/close)
- **Testing & Results:**
 - Successfully transmitted real-time water level to cloud.
 - Gate control via web was reliable (98% success) despite some SMS delays due to signal.
 - ISO 1926 usability metrics:
 - Functionality:  100%
 - Reliability:  99%
 - Usability:  66.75%
 - Efficiency:  Avg. 650.8 ms load time

Identified Technological Gaps

Area	Given Paper	My Proposed Smart System	Technological Gap
Control Intelligence	Manual SMS/web-based gate control based on observed water level	AI/ML predicts future irrigation need based on multiple sensors	No predictive control or intelligent scheduling ; system is fully reactive
Rain Forecasting	Not integrated; decisions made solely on current water level	Integrates weather forecast/rain prediction to avoid unnecessary irrigation	Lacks environmental data fusion for optimal water usage
Irrigation Need Estimation	Gate controlled when level exceeds 5 cm; no moisture or evapotranspiration logic	Predicts how much water is needed and when based on soil + weather + crop	No estimation of moisture requirement; cannot adapt to crop needs

Automation Level	Partially automated via SMS/web buttons	Fully autonomous system that closes loop from sensor → prediction → actuation	Still dependent on human input for triggering actions
Scalability	Designed for single village (Bali)	Designed for wider paddy deployments in Sri Lanka	Not scalable or auto-learning

Value Addition in My Smart Water Gate FYP

- Uses **intelligent AI models** to predict moisture requirements instead of relying solely on current water level.
- Supports **proactive irrigation decisions**, integrating **real-time soil moisture + weather forecasts**.
- Enables **closed-loop automation** where predicted needs directly trigger **motorized gate control**, removing human dependency.
- Allows for **dynamic, crop-sensitive scheduling**, adapting to field variability and optimizing both yield and water use.
- Maintains the paper's practical strengths (web/SIM control) while extending functionality to **intelligent, fully autonomous operation**.

Research Paper 16 - Design and Control System of Sluice Gate With Web-Based Information

Original Research Paper Link -

https://www.researchgate.net/publication/354999767_Autogation_An_Alternate_Wetting_and_Drying-Based_Automatic_Irrigation_and_Paddy_Water_Level_Control_System_through_Internet_of_Things

My Google Drive Link –

https://drive.google.com/file/d/1rSFFPjeLrgbYDUby27PAUW6DvfGTr0-g/view?usp=drive_link

Summary

- **Authors:** Tolentino et al.
- **Published In:** *AGRIVITA Journal of Agricultural Science*, 2021
- **Focus:**

- Developed an **IoT-based automated irrigation system** using the **Alternate Wetting and Drying (AWD)** method to optimize water use in paddy fields.
- Utilizes **ultrasonic water level sensors, Arduino Mega, stepper motor-driven delivery gates, and Zigbee-based wireless transmission.**
- Aims to conserve water by maintaining the paddy water level between **100 mm below ground (dry threshold)** and **150 mm above ground (wet threshold).**
- **System Features:**
 - Real-time water level sensing using a **buried PVC-encased ultrasonic sensor.**
 - Web dashboard provides monitoring of water levels, irrigation status, and gate state (open/close).
 - Uses **Zigbee + Wi-Fi + solar power** for rural deployment and wireless data transmission.
 - Open-source microcontrollers control a **counterweight-designed delivery gate.**
 - Field-tested in Pampanga, Philippines, saving **~20% water over 3 weeks** on a 2-hectare rice field.
- **Conclusion:**
 - The system significantly improves irrigation efficiency using a fixed AWD rule.
 - However, it lacks **predictive logic**, weather integration, or AI-based adaptive control.

Identified Technological Gaps

Area	Given Paper	My Proposed Smart System	Technological Gap
Decision Strategy	Uses fixed AWD logic (100 mm below & 150 mm above) with reactive gate control	Predicts future irrigation needs using soil moisture forecasting & weather prediction	No adaptive or learning-based decision-making model
Rain Forecast Integration	No rain forecast API or prediction model used	Includes weather API or ML-based rainfall forecasting	Cannot avoid irrigation before rainfall; water may be wasted
Crop-Specific Scheduling	Fixed cycle based on water level only	Adjusts irrigation logic based on crop stage + evapotranspiration + soil data	No dynamic crop-stage adaptation
Automation Intelligence	Works on threshold logic with remote sensor + actuator feedback	Uses AI/ML models to control water release based on predicted needs	Lacks predictive intelligence, adaptive scheduling, or optimization
Crop Relevance	General use	Designed specifically for paddy fields with adaptive modes	Not crop-specific or stage-specific

Value Addition in My Smart Water Gate FYP

- **Integrates ML models for predictive soil moisture estimation**, making irrigation smarter and proactive.
- **Includes weather-based decision-making**, reducing unnecessary water use when rainfall is expected.
- Enhances AWD logic with **adaptive scheduling**, dynamically adjusting to field, weather, and crop conditions.
- Delivers a **fully automated control loop** from sensors → prediction → gate control, removing human dependency.
- Supports **field-wide, scalable deployments** using modular, low-cost components with real-time cloud dashboards.
- Enabling **crop-specific water logic** tailored to paddy cultivation

Research Paper 17 - Water Management and Hydrological Characteristics of Paddy Fields

Original Research Paper Link -

https://www.researchgate.net/publication/381828405_Water_Management_and_Hydrological_Characteristics_of_Paddy-Rice_Fields_under_Alternate_Wetting_and_Drying_Irrigation_Practice_as_Climate_Smart_Practice_A_Review

My Google Drive Link –

https://drive.google.com/file/d/10tW4zIQghzTXwzyUqULRQcTCEfiEckaZ/view?usp=drive_link

Summary

- **Authors:** Aziz, Yaacob, Hussain
- **Published In:** *IOP Conference Series: Earth and Environmental Science*, 2021
- **Focus:**
 - Evaluates the effectiveness of the **Alternate Wetting and Drying (AWD)** method for water management in paddy cultivation.
 - Compares AWD with **continuous flooding (CF)** using two experimental blocks in a 1200 m² field.
- **Methodology:**
 - AWD: Water allowed to drop to 15 cm below the surface before re-irrigation.
 - CF: Constant water level maintained throughout the season.
 - Parameters monitored:

- **Water table depth**
- **Soil moisture at 15 cm and 30 cm**
- **Water input and rainfall**
- Data collected using **perforated water tubes**, moisture sensors, and manual water logging.
- **Findings:**
 - **AWD reduced water usage by ~29.2%** compared to CF.
 - Soil moisture under AWD was stable and did not drop below safe crop levels.
 - Water table in AWD fluctuated between -5 cm to -20 cm, versus 0 cm in CF.
 - No adverse effects on plant growth reported.
- **Conclusion:**
 - AWD is an effective water-saving method.
 - However, the system is **entirely manual** — no automation, sensors, or predictive scheduling are included.

Identified Technological Gaps

Area	Given Paper	My Proposed Smart System	Technological Gap
Control Mechanism	Manual tracking and flooding; irrigation done by field staff	Automated gate actuation using real-time sensor and AI prediction	No automation or remote execution; fully manual
Irrigation Scheduling	Based on visual monitoring and fixed AWD threshold (15 cm below surface)	Predicts exact water need based on sensor fusion (moisture, forecast, etc.)	No forecasting or intelligent scheduling capability
Rainfall Prediction	Rainfall only recorded manually; not used to delay or skip irrigation	Includes weather API/rain prediction to optimize timing	Misses opportunity to conserve water during natural rainfall
Data Source & Feedback	Uses PVC tubes and manual sensor readings; no dynamic data loop	Live IoT-based sensors for real-time feedback and prediction	Data not used dynamically to adjust irrigation decisions

Value Addition in My Smart Water Gate FYP

- Automates the proven AWD technique using **sensor-driven water level and moisture monitoring**.
- Adds **AI-powered prediction and weather integration** to skip or delay irrigation when rain is expected.
- Replaces manual irrigation with **microcontroller-based motorized gates**, ensuring accurate and timely water release.
- Transforms field data into **actionable control decisions**, enabling **scalable deployment** across farms without human intervention.

- Enables **smart, adaptive irrigation schedules**, improving water conservation and reducing labor.

Research Paper 18 - Prediction of Rice Water Requirement Using FAO-CROPWAT-8.0 Model in Dehradun, Uttarakhand, India

Original Research Paper Link -

https://www.researchgate.net/publication/340302993_Prediction_of_Rice_Water_Requirement_Using_FAO-CROPWAT-80_Model_in_Dehradun_Uttarakhand_India

My Google Drive Link –

https://drive.google.com/file/d/1N1jW-dN_B98mC-QuHPlpwmRAJ12Tz9sq/view?usp=drive_link

Summary

- **Authors:** Vikash Singh & Ramesh Verma
- **Published In:** *International Journal of Current Microbiology and Applied Sciences*, 2020
- **Focus:**
 - Estimates **water and irrigation requirements** for rice in Dehradun using the **FAO CROPWAT-8.0 model**.
 - Utilizes **crop coefficient method**, **FAO Penman-Monteith equation**, and **USDA-SCS method** for rainfall effectiveness.
- **Methodology:**
 - Collected meteorological parameters (temperature, humidity, wind speed, solar radiation).
 - Calculated **reference evapotranspiration (ET_o)**.
 - Simulated monthly **effective rainfall** and **crop evapotranspiration (ET_{crop})**.
 - Determined **net irrigation requirement** in mm/month and MCM for the region.
- **Results:**
 - Peak rice water demand: **June (8.9 MCM)** and **July (11.2 MCM)**.
 - Maximum ET_o: May (5.28 mm/day); Minimum: January (1.43 mm/day).
 - AWD (Alternate Wetting and Drying) not considered; purely planning-based.
- **Conclusion:**
 - Model helps design region-specific irrigation schedules to improve water use efficiency.
 - However, the system is **manual, simulation-based**, and not linked to any real-time sensor or automated control logic.

Identified Technological Gaps

Area	Given Paper	My Proposed Smart System	Technological Gap
Irrigation Planning	Uses monthly averages via CROPWAT simulation	Predicts real-time irrigation need using sensors + ML	Lacks live adaptability; offline estimation only
Forecasting and Responsiveness	Static evapotranspiration-based schedule	Rain forecast + soil sensor fusion used to skip/delay irrigation	Cannot adapt to real-time weather changes
Execution & Automation	Planning tool for humans to manually irrigate	Microcontroller-based gate control automates irrigation	No automation; only gives irrigation estimates
Data Source	Historical climate data + crop coefficients	Real-time IoT sensors and weather API input	No real-time field data or adaptive decision-making
Forecast Use	Uses historical climate data only	Includes upcoming weather forecasts for intelligent timing	No forward-looking decisions

Value Addition in My Smart Water Gate FYP

- Transforms static, seasonal simulations into **real-time predictive irrigation management**.
- Replaces manual irrigation scheduling with **automated gate actuation** using microcontrollers and live sensor input.
- Enhances water use efficiency by **adjusting in real-time to rainfall forecasts, water tank level and soil moisture predictions**.
- Deploys a **scalable, sensor-integrated field system**, applicable beyond theoretical modeling.
- Considers **rainfall, temperature, soil type, tank level** to optimize gate timing

Research Paper 19 - IoT-based Smart Irrigation System using Machine Learning for Agricultural Water Management

Original Research Paper Link -

https://arxiv.org/abs/2301.00899?utm_source=chatgpt.com

My Google Drive Link –

https://drive.google.com/file/d/19KdhzNfy0dCoY0UtdGuNUHA3kE4b2yyM/view?usp=drive_link

Summary

- **Authors:** Yuji Saikai, Gregory Peake, Karine Chenu
- **Published In:** arXiv (2023)
- **Focus:**
 - Proposes a **Deep Reinforcement Learning (DRL)**-based system for **optimizing irrigation scheduling** in wheat farming.
 - The model learns to make **irrigation decisions daily** using **high-dimensional sensor feedback** and is trained entirely within the **APSIM-Wheat crop simulator** environment.
- **Machine Learning Model Description:**
 - The irrigation problem is modeled as a **Markov Decision Process (MDP)**.
 - The **agent (policy network)** receives a **9-dimensional state vector** as input, which includes:
 - Crop stage
 - Leaf area index (LAI)
 - Extractable soil water from 5 different soil layers
 - Cumulative rainfall
 - Cumulative past irrigation
 - The **action space** is discrete, allowing the agent to choose one of **five irrigation levels** daily (0 mm, 10 mm, 20 mm, 30 mm, or 40 mm).
 - The **reward function** maximizes **profit**:

$$\text{Reward} = \text{Grain Revenue} - \text{Irrigation Cost} \quad \text{Reward} = \text{Grain Revenue} - \text{Irrigation Cost}$$

- The DRL algorithm used is **REINFORCE**, a **policy gradient method**:
 - A **fully connected neural network** (1.4 million parameters) serves as the policy model.
 - The network outputs a probability distribution over possible irrigation actions.
 - The model is trained over **20,000 episodes**, each simulating a full crop season.
- Evaluation was conducted on 10 years of **unseen weather data (2011–2020)**.
 - The DRL policy consistently outperformed the conventional rule-based irrigation strategy.
 - In some years, it yielded up to **17% more profit** due to better water-use efficiency and yield outcomes.
- **Conclusion:**
 - The model demonstrates that **deep reinforcement learning can learn adaptive irrigation strategies** directly from simulation data.

- However, the system operates entirely in a **virtual environment** and is not yet connected to **real-world sensors or automation hardware**.

Identified Technological Gaps

Area	Given Paper	My Proposed Smart System	Technological Gap
Application Domain	Simulated wheat fields in Australia using APSIM	Real-world paddy cultivation with dynamic water gates	Not applied to real-time gate control or water release in field irrigation
Execution Mechanism	DRL produces irrigation recommendations in mm/day; no actuator system	Direct control of sluice gates based on predicted water needs	Does not link predictions to physical automation infrastructure
Sensor Usage	Uses virtual sensors in a simulator (e.g., extractable soil water, LAI)	Uses physical soil and water level sensors in field	Relies on simulator inputs; lacks hardware-level integration
Forecast Integration	Uses historical weather, no real-time API forecast in loop	Integrates real-time weather forecasts/rainfall predictions	Not dynamic to real-world sudden rain or dry periods
System Type	Optimization and decision-support system only	End-to-end autonomous control system	No deployment framework for full field automation

Value Addition in My Smart Water Gate FYP

- Bridges the gap between **AI prediction and real-time actuation** by automating gate control based on soil moisture forecasts.
- Extends DRL-driven insights from simulation into a **real-world operational context**, with water level sensors, gate motors, and weather APIs.
- Implements **closed-loop control** from sensing → prediction → gate actuation, enabling intelligent and adaptive irrigation scheduling.
- Designed for **paddy cultivation**, addressing a different crop and hydrological requirement than simulated dryland wheat farming.
- Supports **dynamic scheduling** in response to actual and forecasted rainfall, reducing water waste and human intervention.

Research Paper 20 - Advanced Prediction of Crop Water Requirements Using Machine Learning Models

Original Research Paper Link -

https://eprint.ijisrt.org/id/eprint/357/?utm_source=chatgpt.com

My Google Drive Link –

https://drive.google.com/file/d/1MmkRJ7Df9nmYuVZ8EAqNpeoPo0o4Nk1a/view?usp=drive_link

Summary

- **Authors:** Manikumari Nagappan, Swetha Jayamurugan, Aarthika Kudiarasumani
- **Published In:** *International Journal of Innovative Science and Research Technology*, 2025
- **Focus:**
 - Develops machine learning models to predict **crop water requirements (CWR)** by estimating **Evapotranspiration (ET_o)** using meteorological parameters.
 - Emphasizes optimizing irrigation efficiency and promoting sustainable agriculture.
- **Machine Learning Models Used:**
 - **Long Short-Term Memory (LSTM):**
 - A deep learning model ideal for time-series prediction.
 - Captures long-term temporal dependencies in weather data.
 - LSTM outperformed other models in accuracy: **RMSE = 0.45 mm/day**, **R² = 0.92**.
 - **Support Vector Machine (SVM):**
 - Effective for non-linear regression using RBF kernels.
 - Slightly less accurate: **RMSE = 0.58 mm/day**, **R² = 0.89**.
 - Other models like ANN, RF, and GBM were tested and ranked based on feature importance and performance.
- **Data Sources:**
 - Meteorological data from IMD, NASA POWER, FAO CLIMWAT, and local stations.
 - Parameters: Temperature, humidity, solar radiation, rainfall, wind speed, sunshine hours.
- **Preprocessing Techniques:**
 - Normalization (Min-Max/Z-score), KNN imputation for missing values, PCA for dimensionality reduction.
 - Feature importance determined using Random Forest and GBM rankings.

- **Deployment:**
 - A **Flask-based web application** allows users (e.g., farmers) to input weather data and receive real-time CWR predictions.
- **Use Case:**
 - For maize in Tamil Nadu with given conditions, the system predicted a **CWR of 3.69 mm/day**.
- **Conclusion:**
 - ML models, particularly LSTM, can significantly improve CWR prediction accuracy.
 - Future work includes integrating IoT sensors, hybrid models, explainable AI, and mobile/web applications for widespread adoption.

Identified Technological Gaps

Area	Given Paper	My Proposed Smart System	Technological Gap
Execution & Automation	Web app provides CWR prediction; manual execution required	Fully automates gate control via microcontroller	No real-time gate actuation or automated irrigation
Real-Time Integration	Relies on user-entered weather data	Uses live sensor + forecast API for autonomous decision-making	No sensor-to-actuator pipeline for closed-loop irrigation
Irrigation Scheduling	Provides predicted ET _o or CWR; irrigation left to user	Calculates and executes required irrigation volume	No control logic tied to predicted values
Crop Adaptability	Focused on maize; fixed K _c values	Dynamic adaptation based on crop type, soil, weather, and growth stage	Static coefficients and fixed estimation logic limit flexibility
Crop Scope	General (tested for maize)	Tailored for paddy cultivation	No paddy-specific modeling

Value Addition in My Smart Water Gate FYP

- **Closes the loop from prediction to actuation** by linking ML-based CWR estimates with real-time gate control.
- Uses **sensor fusion (soil, tank, weather)** and ML predictions to **automate irrigation volume and timing**.
- Enhances model usability by removing reliance on manual web inputs; replaces them with **live environmental sensing**.
- Offers **scalable field deployment**, combining accurate ML prediction with autonomous motor control.

- Supports **forecast-driven, field-specific, crop-sensitive** irrigation planning, leading to optimized water use and yield.
- Tailored for **paddy fields**, not just any crop.

Research Paper 21 - Proper Predictions of the Water Fate in Agricultural Lands: Indispensable Condition for Better Crop Water Requirements Estimates

Original Research Paper Link -

<https://www.sciencedirect.com/science/article/pii/S2090447921000678>

My Google Drive Link –

https://drive.google.com/file/d/1ee0iXyMEcmzzPTYA2x6Zf1yXRyaQ2LN/view?usp=drive_link

Summary

- **Authors:** Ahmed Awad, Luo Wan, Mustafa El-Rawy, Mohamed Galal Eltarabily
- **Published In:** Ain Shams Engineering Journal, 2021
- **Focus:**
 - Investigates how the **accuracy of predicting water fate** in farmland affects **Crop Water Requirement (CWR)** estimation.
 - Compares empirical vs. hydrology-based models using:
 - **CROPWAT:** Empirical model using fixed formulas for effective rainfall.
 - **DRAINMOD:** Hydrology simulator that models infiltration, runoff, and drainage flux under real field conditions.
- **Proposed Method:**
 - Introduces a **new approach** using **DRAINMOD's drainage simulations** to determine CWR:
 - If daily **drainage flux** > 0 , the soil is above field capacity → **No irrigation needed**.
 - If **no flux**, $CWR = ET - \text{infiltration}$.
 - In controlled drainage conditions, the system also considers water table vs. field capacity.
- **Testing & Results:**
 - Site: 12 ha wheat land in Yangtze River, China.
 - CROPWAT estimated CWR as **30,840 m³**; the new DRAINMOD approach gave **12,750 m³** (a ~59% reduction).

- Field scenarios tested: varying drainage practices, pipe spacing, and hydraulic conductivity.
- Changes in water fate caused **CWR variation up to 79%**.
- Empirical models (e.g., CROPWAT) could not adapt to such field changes.
- **Conclusion:**
 - **Accurate hydrological modeling is crucial** for precise irrigation planning.
 - The proposed method improves CWR estimation and supports sustainable water use.

Identified Technological Gaps

Area	Given Paper	My Proposed Smart System	Technological Gap
Execution Method	Simulation-based offline approach using DRAINMOD	Fully automated system with real-time prediction + gate control	No real-time irrigation execution
Sensor & Forecast Integration	Uses weather data only for simulation; no live updates or API integration	Uses real-time sensors + forecast data (rain, evapotranspiration)	Not adaptive to real-time changes in field conditions
Predictive Logic	Relies on simulation logic (drainage = no irrigation)	Uses ML models to predict exact future moisture requirement	No AI-based forecasting or data-driven optimization
Automation Level	Manual modeling and interpretation	Autonomous decision-making linked to motorized gate actuation	No actuation or control layer
Target Crop & Field	Focused on wheat in drained temperate fields	Tailored for paddy cultivation , where continuous water management is critical	Does not address paddy-specific needs or standing water management

Value Addition in My Smart Water Gate FYP

- Converts accurate water fate modeling into a **real-time, ML-driven decision-making system** that triggers irrigation.
- Integrates **live soil moisture, rainfall, and water level sensors** to make dynamic, location-specific predictions.
- Adds an **execution layer** via microcontroller-operated sluice gates, closing the loop from prediction to action.
- Makes the system **adaptive and scalable** for different field scenarios and crop types without requiring full hydrological simulations.
- Bridges simulation and execution, ensuring water is delivered when and where needed based on **data-driven insight**.
- Tailored for **paddy fields**, not just any crop.

Conclusion

After a comprehensive review of 21 leading research papers in the areas of soil moisture prediction, crop water requirement estimation, and smart water gate control systems, it is evident that **none of the existing works offer a complete, real-time, predictive, and autonomous solution tailored for paddy irrigation.**

While various papers demonstrate advancements in:

- **Soil moisture prediction using ML/ANN/LSTM,**
- **Water management models like FAO-CROPWAT, DRAINMOD, and APSIM,**
- **IoT-based gate control using sensors, fuzzy logic, PID, or GSM,**

each study falls short in one or more of the following aspects:

- No system closes the loop from **prediction to actuation** (i.e., from moisture/rain prediction to actual gate movement).
- Limited or no integration of **AI-driven irrigation scheduling** (most use rule-based, threshold-based, or simulation-only logic).
- Absence of **real-time decision-making** based on dynamic field data (e.g., tank water level, soil moisture trends, rainfall forecasts).
- Most are tailored for **general or dryland crops**, and fail to address the **unique hydrological demands of paddy cultivation.**

Innovation of My Proposed FYP

My proposed Smart Water Gate System introduces a **novel, integrated solution** by:

- Combining **real-time environmental sensing**(e.g.- water level , soil moisture) , **weather forecasting**, and **AI-powered predictive modeling.**
- Automating **motorized gate control** based on forecasted soil moisture needs — a fully closed-loop system.
- Specifically optimizing for **paddy fields**, where water needs vary significantly by phase and waterlogging must be precisely managed.
- Supporting scalability, modular deployment, and **data-driven adaptability** over time.